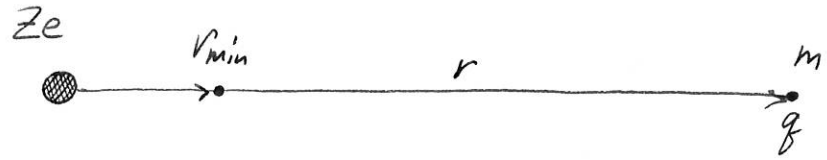
Problem 1



a) The instantaneous energy loss dE_{rad}/dt is given by Eq. (10.77)

$$\frac{dE_{rad}}{dt} = \frac{\mu_0}{6\pi c}\ddot{\vec{d}}^{\,2} = \frac{\mu_0 q^2}{6\pi c} a^2 \qquad (1)$$

The acceleration a is determined from the Newton's law

$$ma = -\frac{dU}{dr}$$

We can now transform the left-hand side of Eq. (1) as

$$\frac{dE_{rad}}{dt} = \frac{dE_{rad}}{dr}\frac{dr}{dt},$$

so that the total radiation on the approach from $r=\infty$ to $r=r_{min}$ is

$$E_{rad}^{(approach)} = \int_{\infty}^{r_{min}} \frac{dE_{rad}}{dr}\,dr = \int_{\infty}^{r_{min}} dr\, \frac{\mu_0 q^2 \left(\frac{dU}{dr}\right)^2}{6\pi c\,(dr/dt)\,m^2}$$

From the conservation of energy

$$\frac{m}{2}\left(\frac{dr}{dt}\right)^2 = \frac{mv_0^2}{2} - U(r) \Rightarrow \frac{dr}{dt} = -\sqrt{v_0^2 - \frac{2U(r)}{m}}$$

so that $$E_{rad}^{(approach)} = \frac{\mu_0 q^2}{6\pi c m^2}\int_{r_{min}}^{\infty} dr\, \frac{(dU/dr)^2}{\sqrt{v_0^2 - 2U(r)/m}}$$

Note that $$v_0^2 - \frac{2U(r_{min})}{m} = 0$$

The energy radiated during departure is equal to $E_{rad}^{(approach)}$, so that the total loss

$$E_{rad} = \frac{\mu_0 q^2}{3\pi c m^2} \int_{r_{min}}^{\infty} dr \frac{(dU/dr)^2}{\sqrt{v_0^2 - 2U(r)/m}}$$

This formula is valid as long as $E_{rad} \ll m v_0^2$.

b) $\frac{dU}{dr} = -\frac{zeq}{4\pi\varepsilon_0 r^2}$ and Eq. (2) gives

$$E_{rad} = \frac{\mu_0 q^2}{3\pi c m^2} \frac{z^2 e^2 q^2}{16\pi^2 \varepsilon_0^2} \int_{r_{min}}^{\infty} \frac{dr}{r^4 \sqrt{v_0^2 - \frac{zeq}{2\pi\varepsilon_0 m r}}} =$$

$$= \frac{\mu_0 z^2 e^2 q^4}{48\pi^3 c m^2 \varepsilon_0^2 v_0} \int_{r_{min}}^{\infty} \frac{dr}{r^{7/2}\sqrt{r - r_{min}}}$$

where $r_{min} = \frac{zeq}{2\pi\varepsilon_0 m v_0^2}$

The integral can be easily rescaled by the substitution: $r = r_{min} y$

$$\int_{r_{min}}^{\infty} \frac{dr}{r^{7/2}\sqrt{r - r_{min}}} = \frac{1}{r_{min}^3} \int_1^{\infty} \frac{dy}{y^{7/2}\sqrt{y-1}} = \frac{16}{15} \frac{1}{r_{min}^3}$$

$$E_{rad} = \frac{\mu_0 z^2 e^2 q^4}{45\pi^3 c m^2 \varepsilon_0^2 v_0} \frac{(2\pi\varepsilon_0 m v_0^2)^3}{(zeq)^3} = \frac{8\mu_0 \varepsilon_0 m v_0^5 q}{45 z e c}$$

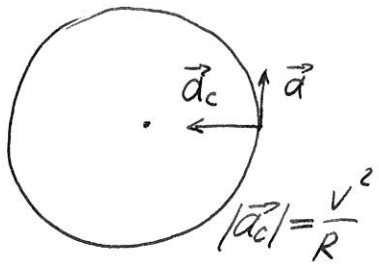
Recalling now that $\mu_0 \varepsilon_0 = \frac{1}{c^2}$ and $E_0 = \frac{m v_0^2}{2}$ we find

$$\frac{E_{rad}}{E_0} = \frac{16}{45}\left(\frac{q}{ze}\right)\left(\frac{v_0}{c}\right)^3$$

We observe that, since $v_0 \ll c$, the losses are small, $E_{rad} \ll E_0$.

Problem 2 Electric dipole radiation

$$\frac{d\mathcal{E}_d}{dt} = \frac{\mu_0 q^2 \vec{a}^2}{6\pi c}$$



The acceleration consists of a tangential component (constant in time) and a centripetal one:

$$\vec{a}^2 = a^2 + \frac{v^4}{R^2} = a^2\left(1 + \frac{a^2 t^4}{R^2}\right)$$

N revolutions take time t_N to perform:

$$\frac{a t_N^2}{2} = 2\pi R N \implies t_N = \sqrt{\frac{4\pi R N}{a}}$$

The amount of energy radiated from $t=0$ to $t=t_N$:

$$\mathcal{E}_N^{(d)} = \frac{\mu_0 q^2}{6\pi c}\int_0^{t_N} \vec{a}^2(t)\,dt = \frac{\mu_0 q^2 a^2}{6\pi c}\int_0^{t_N} dt\left(1 + \frac{a^2 t^4}{R^2}\right) =$$

$$= \frac{\mu_0 q^2 a^2}{6\pi c}\left(t_N + \frac{a^2 t_N^5}{R^2}\right) = \frac{\mu_0 q^2 a^2}{6\pi c}\sqrt{\frac{4\pi R N}{a}}\left(1 + \frac{16\pi^2 R^2 N^2 a^2}{a^2 R^2}\right)$$

$$= \frac{\mu_0 q^2 a^{3/2}\sqrt{RN}}{3\sqrt{\pi}\, c}\left(1 + 16\pi^2 N^2\right)$$

Magnetic dipole radiation

$$\frac{d\mathcal{E}_m}{dt} = \frac{\mu_0}{6\pi c^3}\ddot{\vec{m}}^2 \quad , \text{ where } \quad \vec{m} = \frac{q}{2}\vec{r}\times\vec{v}$$

$$\vec{r} = R\hat{x}\cos\frac{at^2}{2R} + R\hat{y}\sin\frac{at^2}{2R}$$

$$\vec{v} = \dot{\vec{r}} = -\hat{x}\,at\sin\frac{at^2}{2R} + \hat{y}\,at\cos\frac{at^2}{2R}$$

$$\vec{r}\times\vec{v} = Rat\hat{z}\cos^2\left(\frac{at}{2R}\right) + Rat\hat{z}\sin^2\left(\frac{at}{2R}\right) = Rat\hat{z}$$

Since $\vec{m}(t) \propto t \Rightarrow \ddot{\vec{m}} = 0$ and the magnetic dipole radiation is absent.

Problem 3

Equation of motion:

$$m\ddot{x} = -kx + \frac{\mu_0}{6\pi c} q^2 \dddot{x}$$

We look for a solution in the form

$$x \sim e^{-i\omega t}$$

$$-m\omega^2 = -m\omega_0^2 + i\frac{\mu_0 q^2 \omega^3}{6\pi c}, \quad \text{where } \omega_0^2 = \frac{k}{m}$$

$$\omega^2 = \omega_0^2 - i\frac{\mu_0 q^2 \omega^3}{6\pi c m}$$

Treating the last term as a small correction $\omega = \omega_0 - i\gamma$, we get

$$\omega_0^2 - 2i\gamma\omega_0 = -i\frac{\mu_0 q^2 \omega_0^3}{6\pi c m} + \omega_0^2 \Rightarrow \gamma = \frac{\mu_0 q^2 \omega_0^2}{12\pi c m}$$

The second solution is found by setting

$$\omega = -\omega_0 - i\gamma$$

$$x(t) = e^{-\gamma t}\left(Ae^{-i\omega_0 t} + Be^{i\omega_0 t}\right)$$

$|x(t)|^2 \sim e^{-2\gamma t}$, hence after time

$$\tau = \frac{1}{2\gamma} = \frac{6\pi c m}{\mu_0 q^2 \omega^2}$$ the energy decreases by 'e' times.

(over time $\frac{\ln 2}{2\gamma}$ by a factor 2)